Choosing Statistical Techniques, Statistical Pitfalls, Statistics for Machine Learning & Business Dashboard KPIs

Role of Statistics in Decision Making?
Understand how statistical evidence helps businesses make informed decisions.
Choosing the Right Technique:
Learn why the business question determines the appropriate analytical method.

Example:
Investigatinge declining sales requires a different approach from forecasting the next month sales.

Choosing the Right Statistical Technique

1. Discriptive Statistics:
Summarizes data using mean, median, standard deviation and other measures.
Example: Finding average monthly Revenue.
2. Correlation Analysis:
Measures the strength and direction of relationship betweeen two numerical variables. Correlation does not establish causation.
Example: Examining the relationship between ad spend and sales
3. Regression Analysis:
Measures the strength between variables that can support predictions.
Example: Predicting sales using ad spend and product type.
4. Hypothesis Testing:
Uses sample evidence to evaluate a claim about a population.
Example: Testing whther two marketing campaigns have different conversion rates.
5. Confidence Intervals:
Estimates a plausible range for a population parameter using the sample data.
Example: Estimating average customer spendig with a 95% confidence interval.
6. Chi-Square test:
Examines whether two categorical varaiables are associated.
Example: Checking whther product preference is associated with membership type.
7. Time-Series Analysis:
Studies data collected over the time to indentify trends and seasonal patterns.
Example: Analysing monthly sales to identify festive-season demand.
8. A/B testing:
Compare two versions of a product or campaign using defined metrics and statistical methods.
Example: Comparing two wesite checkout designs.

Business Objective	Suitable Technique
Summarize performance	Descriptive statistics
Measure relationships	Correlation
Estimate relationships or predict outcomes	Regression
Evaluate a claim or compare groups	Hypothesis testing
Estimate a population parameter	Confidence interval
Examine categorical associations	Chi-square test
Analyse trends and seasonality	Time-series analysis

Common Statistical Pitfalls

1. Correlation vs Causation

Related variables do not necessarily have a cuase and effect relationship.

Example: Ice -Cream sales and swimming activity may both increase during summer.

2. Sampling Bias

An unrepresentative sample produce misleading conclusions.

Example: Surveying only premium customer to asses across all the customer.

3. Small Sample Size

Small samples may provide imprecise estiamtes and fail to represent the popualtion adequately.

Example: Using feedback from 10 customers to represent 10,000 customers.

4. Outliers

Extreme observations can distort averages and influence regression results. Investigate outliers before removing them.

Example: One unusually large trabsaction increases average order value.

5. Misinterpreting Statistical Significance

A statistical significance result is not nessarily important commercially.

Example: A tiny conversion improvement may have little financial value.

6. Multiple comparisons

Testing many hypothesis inrease the chance of false-positive findings unless appropriate adjustments are considered.

Example: Testing 100 marketing variations may produce apparently significant results by chance.

7. Simpson's Paradox

A relationship observed withing separate groups can change or reverse when the groups are combined.

Example: A treatment may perform better in each customer segment but worse in the combined data because segment sizes differs.

8. Misleading Visualization

Inappropriate scales or missing context can exaggerate differences.

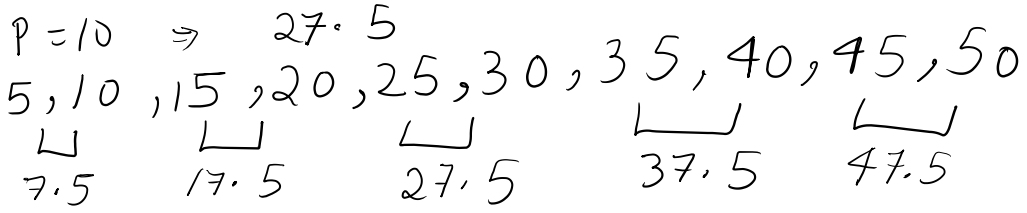
Example: A truncated bar chart can make a small sales increase appear substantial.

Statisitics For Machine Learning

1. Population, Sample and Sample variability

A sample represents a part of population, and different samples produce different esitmates.

Example: Average spending among 500 customers may differ from the population average.



2. Mean, variance and SD

These measures describe the center and variability of data.

Example: Two products may have identical average sales but different sales fluctuations.

3. Probability Distributions

Probability decribes uncertainty, while distributions decribe possible values and their likelihoods

Example: Analysing daily order counts to understand demand variability.

4. Training vs Testing Data

Training data fits the model, testing data evaulates performance on unseen observations.

Example: Training churn model on historical customers and evaluating it on a separate dataset.

5. Overfitting vs Underfitting

Overfitting captures noise or overly specific patterns; underfitting fails to capture important patterns.

Example: A forecasting model performs well on historical data but poorly on future sales.

6. Regression Classification

Regression predicts numerical outcomes, while classification predicts categories.

Example: Prediciting Revenue is regression, predicting whthera customer will churn is classification.

7. Model Evaluation Metrics

Metrics measure different aspects of prective performance

i) MAE: Average absolute prediction error.

ii) RMSE: Penalizes larger prediction errors more heavily

iii) Accuracy: Proportion of correct classifications

iv) Precision and Recall: Evaluate different aspects of positive-class predictions.

8. Data Leakage

Data leakage occurs when information unavailable at prediction time imporperly influences model training or evaluation

Example: Using a customer's cancellation date to predict whther that customer will cancel.

Business Dashboard Metrics & KPIs

1. Metrics vs KPIs

Metrics measures performance; KPIs track progress toward important business objectives.

Example: Website visits are a metric, while conversion rate may be a KPI

2. Revenue, Profit and Profit Margin

Revenue measures sales income, while profit accounts for expenses. Profit margin expresses profit as a percentage or revenue.

$$PM = R - TEx$$

$$PM = (P \div R) \times 100$$

Example: Revenue ₹5,00,000 and expenses ₹4,00,000 produce ₹1,00,000 profit and a 20% profit margin

3. Conversion rate:

Measures the percentage of eligible visitors who complete a desired action.

Conversio rate = (Conversions / Eligible Visitors) x 100

Example: 500 purchases from 10,000 vistors = 5%

4. Customer Acquisition Cost (CAC)

Measure acquisition costs per newly acquired customer.

CAC = Acquisition Costs / New customers acquired

Example: ₹60,000 spent to aquire 300 customers = ₹200 per customer

5. Average order value

Measures average revenue per order

AOV = Total Revenue / Total orders

Example: ₹2,00,000 revenue from 500 orders , AOV = ₹400

6. Retention and Churn

Retention measures the proportion of customers who remain, while churn measures the proportion who leave during a defined period. Use consitent cohort definitions.

7. MoM and YoY Growth

Month over month compares the current month with th previous month; Year over year compares same period across years.

Example: Revenue increase from ₹10 lakh to ₹12 lakh represents 20% growth.

8. Statistical Interpretation Of KPIs

KPI changes may reflect genuie improvements, normal variation, or changes in customer mix. Sample size and context matter when interpreting results.

Case Study: E-Commerce Dashboard

KPI	Previous Month	Current Month
Revenue	10 Lakh	12 Lakh
Orders	2,000	2,400
Website Visitors	20,000	24,000
Marketing Spend	₹ 1,00,000	₹ 1,20,000
	Curr_Mont	Prev Month
Revenue Growth =	20	
Order growth =	20	
Conversion Rate =	10	10
AOV =	500	500
Marketing spend per order	50	50

Business Insight

Revenue and orders increased by 20%, while conversion rate and average order value remained unchanged.
Further cost and profit analysis is needed to determine whther business performance improved overall.

Real Life Business Applications

Case Study: retail Business with Declining Profit Margins

1. Revenue vs Costs:

Determine whether expenses are increasing faster than revenue.

2. Discount Analysis:

Examine the relationship between discounts and profit margins.

3. Category Performance:

Identify which product categories contribute most to revenue and profit

4. Outlier Detection:

Investigate unusual transactions and potential data errors

5. Time-Series analysis:

Examine sales trends and seasonal patterns

6. Satatistical Interpretation:

Determine whther observed changes justify further investigation or a business decision.

Expected Outcome:

Recommend appropriate statistical techinques and support conclusions with calculated KPIs and evidence.